

2) $\phi(5w) = \lg^{-1} \left(\frac{-w^5 + 5w^3 - 2w}{2w^4 - w^2 + 5} \right)$ ^{usamos} componentes pares $\Rightarrow$ no pueden haber polinomios en semi- ϕ o ϕ de grado impar

Planteo la siguiente igualdad con $\$ = 5w$

$$\frac{Z(\$)}{Z(-\$)} = \frac{1 + \$ \lg(\phi(5w))}{1 - \$ \lg(\phi(5w))} = \frac{1 + \$ \cdot \frac{-w^5 + 5w^3 - 2w}{2w^4 - w^2 + 5}}{1 - \$ \cdot \frac{-w^5 + 5w^3 - 2w}{2w^4 - w^2 + 5}}$$

$$= \frac{2w^4 - w^2 + 5 \cdot (-\$w^5 + 5\$w^3 - 2\$w)}{2w^4 - w^2 + 5 \cdot (1 - \$ \cdot \frac{-w^5 + 5w^3 - 2w}{2w^4 - w^2 + 5})} = \frac{-\$w^5 + 5\$w^3 - 2\$w}{2w^4 - w^2 + 5 + \$w^5 - 5\$w^3 + 2\$w}$$

$$= \frac{-\$w^5 + 2w^4 + 5\$w^3 - w^2 - 2\$w + 5}{\$w^5 + 2w^4 - 5\$w^3 - w^2 + 2\$w + 5}$$

$\$ = 5w$ $\lg(5w)^5 = 5 \cdot w^5$
 $5^5 = 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 = 1.5$
 $5^3 = 5 \cdot 5 \cdot 5 = -5$
 $5^2 = 5 \cdot 5 = -1$
 $5^4 = 5 \cdot 5 \cdot 5 \cdot 5 = 1$

$$= \frac{-(5w)^5 + 2(5w)^4 + (5w)^3 \cdot 5 + (-5w)^2 \cdot 2 \cdot 5w + 5}{(5w)^5 + 2(5w)^4 + (5w)^3 \cdot 5 + (-5w)^2 \cdot 2 \cdot 5w + 5}$$

$$= \frac{-\$^5 + 2 \cdot \$^4 - 5\$^3 + \$^2 - 2 \cdot \$ + 5}{\$^5 + 2 \cdot \$^4 + 5\$^3 + \$^2 + 2 \cdot \$ + 5}$$

$$= \frac{(\$ - (-\sqrt[3]{-1})) \cdot (\$ - (-1)^{2/3}) \cdot (\$ - 1) \cdot (\$ - (1 - 25)) \cdot (\$ - (1 + 25))}{(\$ - \sqrt[3]{-1}) \cdot (\$ - (-1)^{1/3}) \cdot (\$ + 1) \cdot (\$ - (-1 - 25)) \cdot (\$ - (-1 + 25))}$$

1	2	5	1	2	5	1	2	-5	1	-2	5		
-1		-1	-1	-4	5	1		-1	1	4	-3	-5	
	1	1	4	-3	5	0		-1	1	-4	-3	-5	0

$$(\$+1)(\$^4 + \$^3 + 4\$^2 - 3\$ + 5)$$

$$(\$+1)(\$^2 + 2\$ + 5)(\$^2 - \$ + 1)$$

$$(\$-1)(-\$^4 + \$^3 - 4\$^2 + 3\$ - 5)$$

$$(\$-1)(\$^2 - 2\$ + 5)(\$^2 + \$ + 1)$$

Finalmente:

$$Z(\$) = \$^2 + \$ + 1$$

esto toma los polinomios del semigrupo $\mathbb{Z}$